

Graph-Aware Anomaly Detection for Blockchain Forensics

A 48-Experiment Benchmark on the Elliptic Bitcoin Dataset

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Why Bitcoin Forensics is Hard

The setting

- Bitcoin blockchain: fully **public** but **pseudonymous**
- Every transaction recorded; wallet identity is hidden
- Money laundering, ransomware, darknet markets
- Traditional AML requires verified identity — unavailable here

Three core challenges

- 1 **Severe imbalance** — only **2%** illicit
Accuracy is useless (93.5% by predicting all-licit)
- 2 **Concept drift** — illicit rate drops
11.6% → 6.5% train→test
- 3 **Anonymous features** — 165 features, semantics unpublished

Primary metric: F1 on illicit class **Secondary:** PR-AUC, ROC-AUC
Accuracy deliberately excluded — misleading under 6.5% positive rate

The Elliptic Dataset

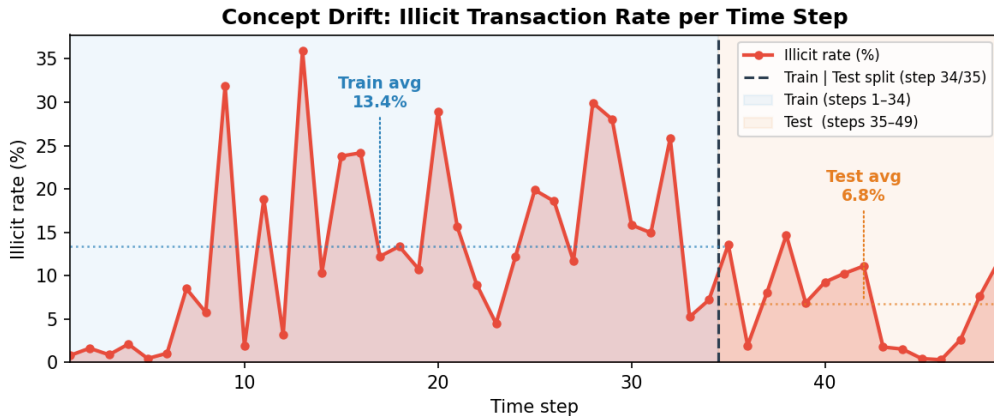
Property	Value
Transactions	203 769
Directed edges	234 355
Avg. node degree	2.30
Illicit (label 1)	4 545 (2%)
Licit (label 2)	42 019 (21%)
Unlabeled	157 205 (77%)
Time steps	49
Local features	93
Aggregated feat.	71

Temporal split (mandatory)

- **Train**: steps 1–34 29 894 labeled rows
11.6% illicit
- **Val**: steps 30–34 (subset of train)
- **Test**: steps 35–49 16 670 labeled rows
6.5% illicit

Random shuffling **leaks the future** into training and inflates every metric. All 48 experiments respect the causal time ordering.

Concept Drift: Illicit Rate Per Time Step



Illicit rate falls **44% relatively** from train to test window. Not noise — genuine behavioural shift requiring temporal-aware evaluation.

Five Experimental Paradigms — 48 Variants, 5 Rounds

① Classical Unsupervised

IsoForest, LOF, OCSVM, spectral

Train on all / licit-only rows; no labels

② Supervised Trees

LR, RF, GBM, LightGBM, Optuna-tuned

Train on 29 894 labeled rows

③ Neural Anomaly Detection

AE, DenoisingAE, VAE, β -VAE, LSTM-AE

Train on 26 432 licit-only rows

④ GNN Supervised

GraphSAGE family, GAT, GATv2

Full graph, node-mask training

⑤ Semi-supervised / Ensemble

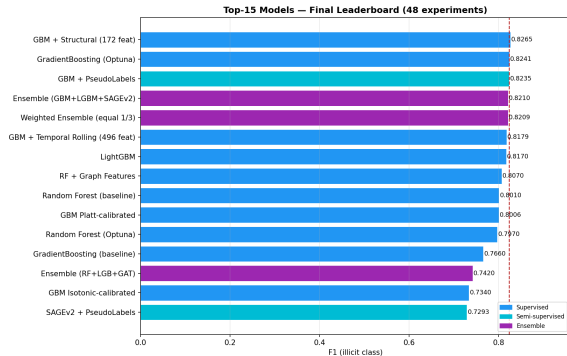
Pseudo-labels, weighted ensemble,

Platt / isotonic calibration

Course-derived methods:

DOMINANT, LSTM-AE, β -VAE, EVT
thresholding, spectral embedding,
pseudo-label propagation

Five Performance Tiers Across 48 Experiments



Tier

F1 range

Supervised / Ensemble 0.74–**0.83**

GNN supervised 0.54–0.73

Neural AE 0.12–0.41

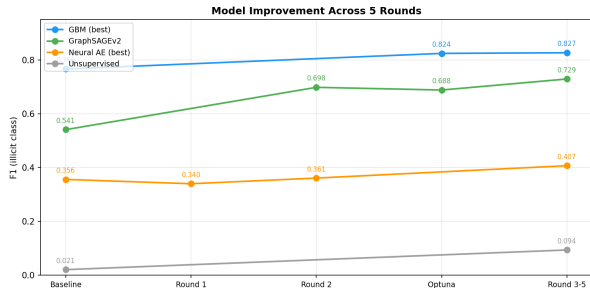
Graph AE (DOMINANT) 0.21

Classical unsupervised <0.10

Key insight

Supervised trees dominate because the 71 aggregated features already encode neighbourhood signal in tabular form. GBM + structural topology: **F1 = 0.8265**

Supervised Results & Improvement Journey

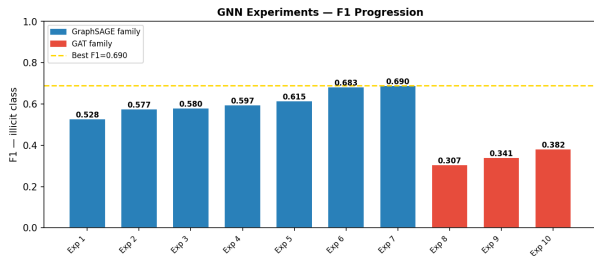


Best supervised results:

Model	F1
GBM + Structural	0.827
GBM Optuna	0.824
LightGBM	0.817
Random Forest	0.801
Logistic Reg.	0.302

- Optuna (+0.058 over GBM baseline)
- Structural topology adds +0.002 on top
- Semi-supervised pseudo-labels:
 $\Delta = -0.0006$ (already saturated)

GNN Ablation: What Actually Matters?



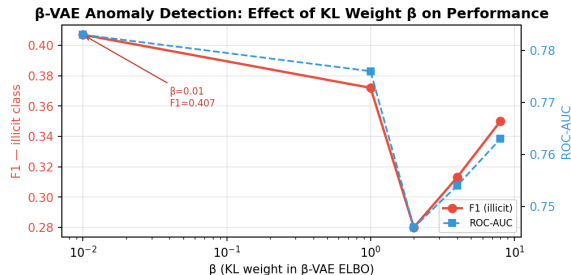
10 controlled experiments:

Design choice	$\Delta F1$
LayerNorm + self-loops	+0.068
Depth (2L $\rightarrow$ 3L)	+0.038
Width (h=64 $\rightarrow$ 256)	+0.052
Optuna lr tuning	+0.007
GAT / GATv2	-0.308 vs SAG

Finding

Attention is noisy on sparse graphs (avg. degree 2.3). Mean aggregation (GraphSAGE) wins.
Best GNN: GraphSAGEv2 **F1 = 0.698**
= **84.5%** of supervised ceiling

Unsupervised Detection: Score Inversion Paradox



Classical unsupervised: all fail

IsoForest F1=0.021, LOF F1=0.076

Score inversion: illicit transactions reconstruct with *lower* MSE than licit ones. Anomaly score must be $-MSE$, not $+MSE$.

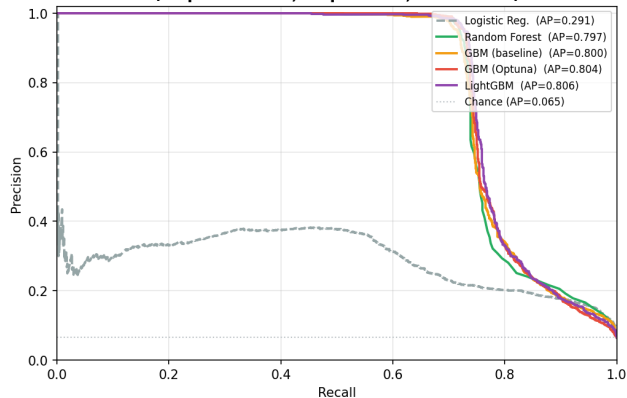
Laundering patterns are **stereotyped**, not rare.

Best unsupervised results:

Model	F1
β -VAE ($\beta=0.01$)	0.407
DenoisingAE	0.361
DOMINANT (graph AE)	0.212
Classical best (LOF)	0.076

Precision–Recall Curves: Supervised Models

Precision–Recall Curves: Key Supervised Models
(Elliptic test set, steps 35–49, 6.5 % illicit)



Model	AP
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LightGBM	0.806
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GBM Optuna	0.804
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GBM baseline	0.800
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Random Forest	0.797
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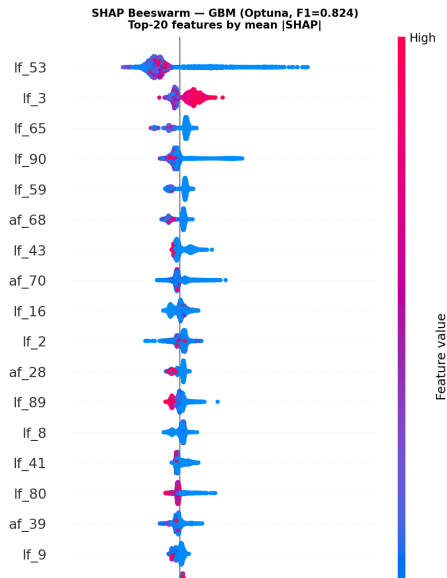
Logistic Reg.	0.291
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Chance line	0.065
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PR-AUC is

threshold-independent and informative under imbalance — it ignores the large pool of true negatives that inflates ROC-AUC.

Cross-Paradigm Feature Agreement



Method:

- SHAP TreeExplainer → GBM
- Gradient attribution → GraphSAGE

Cross-paradigm consensus (top-20 in all models):

lf_53

lf_90

af_70

- lf_* = local transaction features
- af_* = aggregated neighbourhood feature
- af_70 in GNN attribution confirms **graph signal exists** beyond tabular proxies
- Semantics unknown — requires Elliptic12/1

Research Questions — Answered

RQ	Answer
RQ1	Classical unsupervised detectors fail ($F1 < 0.10$). Illicit transactions are stereotyped, not rare — they form dense clusters, not outliers.
RQ2	GraphSAGEv2 recovers 84.5% of the supervised ceiling using only graph message passing. GBM and GNN exploit complementary channels .
RQ3	LayerNorm+self-loops (+0.068) > depth (+0.038) > width > tuning. Attention underperforms mean aggregation on sparse graphs.
RQ4	lf_53, lf_90, af_70 rank top-20 in RF, GBM, and GraphSAGE — robust cross-paradigm signals.

6 key findings:

- 1 **Temporal split is mandatory**
Random shuffling invalidates all metrics
- 2 **Supervised trees dominate**
GBM + structural: **F1 = 0.8265**
- 3 **GNNs add complementary signal**
GraphSAGEv2: **F1 = 0.698** (84.5% of ceiling)
- 4 **AE detection is viable but inverted**
 β -VAE ($\beta=0.01$): F1 = 0.407
- 5 **DOMINANT best unsupervised graph method**
F1 = 0.212, beats all classical by >0.13
- 6 **Robust cross-paradigm features exist**
1f_53, 1f_90, af_70

Future work:

- GBM + GraphSAGEv2 stacking ensemble
- Per-transaction LSTM-AE
- EvolveGCN for temporal drift
- Online threshold recalibration
- GNNExplainer for node-level attribution
- Elliptic collaboration: feature semantics

Bottom line

On Elliptic, illicit transactions are not outliers. Supervised trees + graph structure together provide the strongest forensic signal. Course-derived unsupervised methods (β -VAE, DOMINANT) are viable baselines.

Thank You

Questions?

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Best result: **GBM + Structural, F1 = 0.8265**

Best GNN: GraphSAGEv2, F1 = 0.698

Best unsupervised: β -VAE ($\beta=0.01$), F1 = 0.407

48 experiments · 5 paradigms · strict temporal split